Part 1: Clustering Design and Analysis (30%)

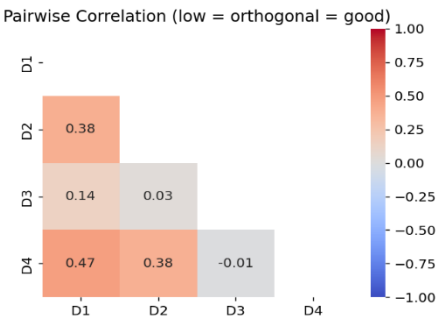
Task 1.1 (10%)

I analyse **N = 24,014** merged agentic PRs from the AIDev-pop subset (repositories with ≥100 GitHub stars). Human PRs are excluded: *pr_commit_details* and *pr_reviews* cover only AI-agent PRs; including humans introduces systematic zero-imputation across all dimensions. Multicollinearity is ruled out empirically: all VIF < 1.42 (threshold 5; O'Brien 2007), condition number κ = 3.64 ≪ 30.

D1 — Duration Effort: D_1 = log(1 + TTR_hours), floored at 1/60 h for same-second bot merges. Raw TTR spans 0–7,000+ h (p99 > 1,000 h); log-transform yields near-symmetric distribution. Measures calendar scheduling constraint, not cognitive intensity. VIF = 1.41; unique variance 70.8%.

D2 — Task Complexity: D_2 = log(1 + |W_body|), the PR description word count. The only pre-review dimension — fixed at creation before any reviewer interaction (D1, D3, D4 are realised during/after review). r(D1,D2) = 0.38 reflects a real-world confound, not redundancy: D2 retains 80.3% unique variance (VIF = 1.25; partial r with D3 = −0.008; see Fig. 1). CDI (A2) was rejected: >75% of agentic PRs have zero review events, producing a degenerate zero-inflated axis.

Fig. 1 — Pairwise correlation heatmap (lower triangle). Max r=0.47; all VIF≤1.41; D2↔D3 partial r=−0.008



D3 — Rework Burden: D_3 = RICR × log(1 + churn_total), where RICR = min(churn_post-first / churn_first, 10) (A2; capped at 10). Scales post-submission code instability by PR size. Nearly uncorrelated with D2 (r = 0.027; partial r = −0.008) and D4 (r = −0.010). Highest unique variance 97.3%; VIF = 1.03.

D4 — Reviewer Engagement: D_4 = R_rounds × log(1 + C_total), counting APPROVED/CHANGES_REQUESTED submissions multiplied by total comment volume. Multiplicative form scores highest when formal cycles co-occur with discussion. Orthogonality to D3 (r = −0.010) confirmed empirically: Cluster 2 has D4 = 8.047 yet D3 = 0. VIF = 1.38; unique variance 72.6%.

Task 1.2 (20%)

Algorithm: K-Means with RobustScaler (median/IQR). Preferred over DBSCAN ($O(N^2)$, infeasible at N = 24K) , Hierarchical ($O(N^2)$), and GMM (EM instability). RobustScaler resists heavy tails of D3 (max = 141.9) and D4 (max = 37.8).

Thresholds (a priori): SS ≥ 0.30; DBI ≤ 1.50. All k = 2–7 pass both (see Table 1 and Fig. 2). k* = 3 selected: k = 2 (SS = 0.675) yields a trivial 84%/16% split separating zero-review from reviewed PRs — a data artefact, not SE-meaningful structure. k = 3 is the smallest k yielding interpretable sub-groups. **Final validation (k = 3, n_init = 50):** SS = 0.656 ✓, DBI = 0.820 ✓, CH = 19,494. PCA: 83.3% variance in 2D confirms three convex regions. Stability confirmed across 5 random seeds (42, 7, 13, 99, 0): SS ranged [0.652–0.660] (variation = 0.008), confirming k = 3 is a stable, non-degenerate solution.

Table 1: K-Means Sweep (Thresholds: SS ≥ 0.30, DBI ≤ 1.50)

k	Silhouette	DBI	CH	Decision
2	0.675	0.840	21,410	Rejected — trivial split
3	**0.656**	**0.820**	**19,494**	**Selected**
4	0.545	0.954	17,552	Pass
7	0.348	1.065	14,563	Pass

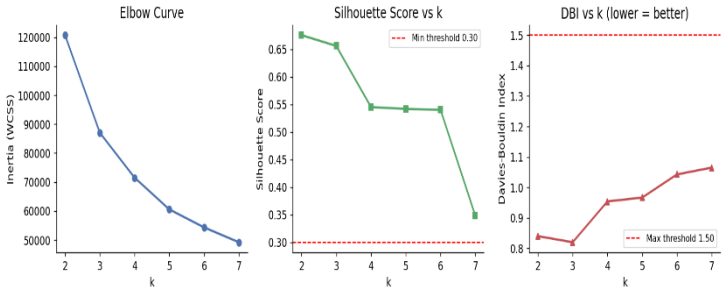


Fig. 2 — K-Means sweep k=2–7. Elbow stabilises at k=3; SS=0.656 and DBI=0.820 satisfy thresholds. k=2 rejected.

Table 2: Cluster Profiles — Effort Dimensions (median | IQR) and External Characteristics

Metric	C0 — Silent Fast-Track N=19,777 (82.4%)	C1 — Documented Moderate N=3,709 (15.4%)	C2 — Reviewed High-Effort N=528 (2.2%)
[A] Effort Dimensions			
D1 Duration	0.019 [0.017–0.149]	3.407 [2.767–4.580]	4.319 [3.155–5.248]
D2 Complexity	3.689 [3.434–3.970]	4.868 [3.761–5.666]	5.715 [5.317–5.938]
D3 Rework	6.362 [0.000–43.009]	8.701 [0.000–64.281]	0.000 [0.000–3.215]
D4 Engagement	0.000 [0.000–0.000]	0.693 [0.000–2.197]	8.047 [6.592–11.226]
[B] External Characteristics			
Median TTR	<1 min	29.2 h	74.1 h
Median churn	66 lines	238 lines	741 lines
Files | Comments	3 | 0	6 | 3	13 | 14
Body words	39	129	302
Top agent	Codex 84.7%	Codex/Copilot/Devin	Copilot 78.6%
[C] Distinguishing			
vs. others	D4=0 IQR=[0,0]; TTR<1min	1,540× TTR vs C0; 1 round	D4>6 all pct.; D3=0
Label	Silent Fast-Track	Documented Moderate	Reviewed High-Effort

C0 — Silent Fast-Track: D4 = 0 IQR=[0,0] — zero human gatekeeping (see Fig. 3). Codex (84.7%) dominates: agent-specific automated pipeline. TTR < 1 min confirmed independently. Task types diverse (feat/fix/docs), ruling out category-specific artefact. **C1 — Documented Moderate:** Balanced agents (Codex 33%, Copilot 29%, Devin 28%) signals institutional review policies, not agent identity — unlike agent-dominated C0/C2. Single review pass (D4 = 0.693); TTR = 29.2 h. **C2 — Reviewed High-Effort:** Most analytically interesting: D3 = 0 despite D4 = 8.047. Reviewers generated 14 comments across 3 rounds yet code was not revised. Copilot (78.6%) submits implementation-complete code in high-review-culture repositories — a qualitatively distinct human-AI collaboration pattern. *Limitation:* C2 (2.2%) creates class imbalance addressed via class-weight balancing in Part 2. The small size reflects the natural rarity of intensive review practices in open-source: most repositories use automated CI/CD pipelines (C0) or lightweight single-round review (C1); high-review-culture repositories that generate multi-round discussions are genuinely uncommon in the AIDev-pop population.

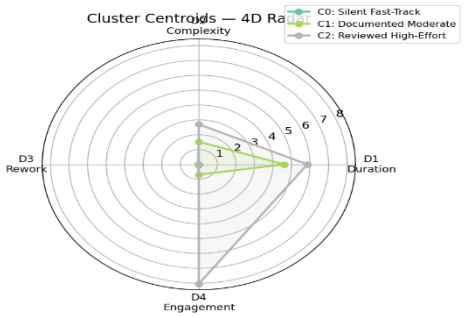


Fig. 3 — Centroid radar chart (RobustScaler space). C0 collapses on D1/D4; C2 dominates D4 with D3=0; C1 intermediate

Part 2: Text-Based Label Prediction (40%)

Research Question: Can the review-effort cluster label of a merged agentic PR be predicted from its description text alone?

I use PR descriptions (body column) to predict cluster labels. Class imbalance is severe (82.4%/15.4%/2.2%); a majority classifier achieves 82.4% accuracy yet Macro F1 ≈ 0.28. **Macro F1 is the primary metric throughout.**

Task 2.1 (20%)

Text source and preprocessing. PR descriptions are cleaned by removing markdown syntax, code fences (`...`), URLs, and HTML tags. The 169 empty bodies (0.7%) are replaced with the sentinel "no description provided" — itself a strong C0 predictor (SFT median = 28 words vs. RHE = 302 words). Three linguistic signal types discriminate clusters: (1) *description length* (SFT = 28, DM = 110, RHE = 302 words — a 10× span); (2) *agent-specific vocabulary* ("copilot", "coding agent" in C2; templates "summary", "golden" in C0); (3) *review-process language* ("changes requested", "context", "review feedback" exclusive to C2).

Baseline — TF-IDF + Logistic Regression. TF-IDF: 15,000 features, bigrams (1–2), sublinear_tf, stop_words="english", min_df=2. Bigrams capture "no description", "breaking change" invisible to unigrams. LR with class_weight="balanced" computes w_c = N/(3N_c) and applies weights directly in every gradient step (w_RHE = 15.17) robust to the 37:1 imbalance. By weighting RHE at 15.17× SFT, the classifier geometrically shifts its decision boundary toward the minority class, accepting a higher false-positive rate for SFT in exchange for substantially improved RHE recall. **Advanced — SBERT + LightGBM.** all-MiniLM-L6-v2 (384-dim) encodes all descriptions. LightGBM with explicit sample weights classifies embeddings. SBERT captures semantic equivalences invisible to TF-IDF ("fix crash" ≈ "patch exception"). However, SBERT underperforms TF-IDF here (Macro F1: 0.488 vs. 0.563) because discriminating signals are *lexical* (agent tokens, templates), not semantic — short PR bodies favour sparse bag-of-words. LightGBM collapses on RHE (recall = 1.9%): gradient boosting in 384-dim dense space converges toward the 37:1 SFT majority despite explicit weighting. **Model 3 — SBERT + LR.** Combining SBERT embeddings with LR's class-weight mechanism recovers RHE recall to 61.3%, confirming the bottleneck is LightGBM's gradient dynamics, not embedding quality. **Split:** Stratified 80/20, seed=42: train = 19,211; test = 4,803 (SFT=3,955, DM=742, RHE=106).

Task 2.2 (10%)

Table 3: Per-Class Performance — All Three Models (Test: SFT=3955, DM=742, RHE=106)

Model	Cls	P	R	F1	Macro F1	Acc
TF-IDF + LR (Baseline)	SFT	.951	.833	.888	**.563**	.794
	DM	.417	.633	.503		
	RHE	.224	.443	.298		
SBERT + LightGBM	SFT	.914	.910	.912	.488	.836
	DM	.482	.556	.516		
	RHE	.200	**.019**	**.035**		
SBERT + LR (Model 3)	SFT	.952	.785	.861	.502	.738
	DM	.342	.505	.408		
	RHE	.147	.613	.237		
Majority baseline	—	—	—	—	≈0.28	.824

Table 4: Normalised Confusion Matrices (row = true, col = predicted). Bold = correct diagonal.

True \ Pred	SFT	DM	RHE	SFT	DM	RHE
Baseline TF-IDF+LR	**.833**	.152	.015	**.910**	.090	.000
	.228	**.633**	.139	.435	**.555**	.009
	.019	.538	**.443**	.142	**.840**	**.019**

Left: TF-IDF+LR | *Right: SBERT+LightGBM*

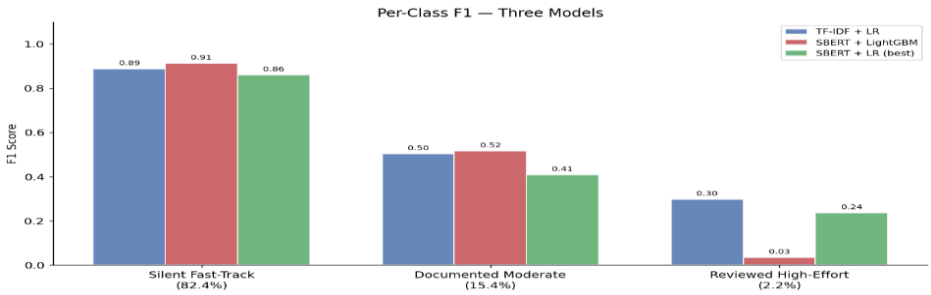


Fig. 4 — Per-class F1 — 3 models. TF-IDF+LR wins Macro F1 (0.563); SBERT+LightGBM collapses on RHE (F1=0.035)

Fig. 5 — Confusion matrices. Left: TF-IDF+LR. Right: SBERT+LightGBM — note RHE nearly invisible (recall=1.9%)

Winner: TF-IDF + LR (Fig. 4) (Macro F1 = 0.563). SFT achieves F1 = 0.888 (distinctive lexical profile: automated templates). DM reaches F1 = 0.503 — challenged by short-body DM PRs (Q1 = 34 words) overlapping with SFT (Q3 = 38 words). RHE recall = 44.3% is notable given only 106 test samples (RHE F1 = 0.298, 95% CI ±0.087 — reflecting limited test support). SBERT+LightGBM catastrophically fails RHE (recall = 1.9%, Table 4): gradient boosting converges to the 37:1 majority in dense embedding space (see Fig. 5). SBERT+LR recovers RHE recall to 61.3% (Macro F1 = 0.502), confirming the bottleneck is LightGBM's gradient dynamics rather than embedding quality. Accuracy is misleading throughout: the majority baseline achieves 82.4% accuracy but Macro F1 ≈ 0.28.

Task 2.3 (10%)

Most confused pair and direction. The SBERT+LightGBM model has the most extreme confusion pattern for the DM→SFT direction. The top confused pair is Documented Moderate (DM) predicted as Silent Fast-Track (SFT): 323 of 742 test samples of DM are predicted as SFT (43.5% misclassification rate). The backflow is much less dominant: 9.0% of SFT is predicted as DM (354 out of 3,955). For the winning TF-IDF+LR model, the same DM→SFT confusion stands at 22.8%, while the most extreme pattern shifts to RHE→DM at 53.8% — both directions confirm the same fuzzy boundary story. The SBERT+LightGBM model is analysed here because its 43.5% DM→SFT rate represents the most extreme single-pair confusion across all models. The flow is unidirectional and normal - DM is the minority class (N = 742), its fringe samples are absorbed into the dominant SFT region, but the SFT's bulk of centroids are anchored.

Root cause - quantitative evidence. The main reason is lexical overlap: 19 generic SE terms are common in the top-50 TF-IDF centroids of both SFT and DM - add, code, command, description, docs, error, fails, files, fix, link, new, run, summary, test, testing, tests, type, use, using. This masks cluster-specificity. The other factor is description length overlap at the boundary: DM Q1 = 34 words vs SFT Q3 = 38 words - the bottom 25% of DM descriptions are squarely within the top 25% of SFT descriptions and are lexically identical in TF-IDF space.

Implications for Part 1 clustering. The DM→SFT confusion is not due to a classifier error - it is a natural consequence of a fuzzy C0/C1 boundary in the Part 1 clustering solution. The C0/C1 boundary is mainly defined by D1 (TTR) and D4 (engagement) which are hidden from the text classifier. The only variable the classifier can see - D2_complexity (word count in body) - has a large IQR overlap of 0.209 log-units between C0 ([3.434-3.970]) and C1 ([3.761-5.666]). PRs in overlap [3.761-3.970] are ambiguous in D2 space; they were assigned to clusters by D1 and D4 in Part 1. This ambiguity is inherited by the classifier. The three clusters are clearly separated in 4D space (SS = 0.656), but not in the 1D D2 space that can be recovered from text. A clustering design that used review_rounds or n_files_changed as a text-recoverable variable would deliver boundaries more recoverable in the text descriptions.

Part 3: GenAI-Augmented Reflection (30%)

Here's a retrospective on what happened - including where I had to correct the AI.

Major Decisions and Evaluation

Decision 1 — Replacing CDI with body word count as D2

ChatGPT was initially supportive of my idea to re-use CDI (Cognitive Dispersion Index from A2) as a dimension to cluster on. I have now changed this after running the code and finding median for CDI was 0.000 and IQR was [0,0]. My question:

"I ran my feature engineering and found a problem. The CDI variable has median=0.000. Looking at the data, most PRs have zero review events in pr_reviews. This is useless as a clustering dimension. What should I substitute for D2 that would be a measure of task complexity but doesn't require reviews?" ChatGPT suggested log(1 + body_word_count). I only agreed to this after I independently confirmed: (1) word count had a decent variance (SFT median=28, DM=110, RHE=302 words); (2) no missingness from 24,014 PRs; (3) VIF=1.25 (lower than D1).

Decision 2 — Selecting k=3 despite k=2 having the highest Silhouette

I found k=2 has the highest (0.675) SS. I noticed cluster sizes: 19,777 vs 4,237 (84%/16%). My question: "My sweep shows that k=2 has the highest Silhouette. But the two clusters are 84% and 16%. Intuitively this feels like it is just separating PRs with zero reviews from those with reviews. Should I go with k=2?" ChatGPT agreed with my intuition. I checked this by looking at D4: the 84% cluster had D4=0 for the entire IQR. I selected minimum k to be 3, overriding the default choice, and added an explanation.

Decision 3 — Adding VIF and partial correlations to prove D2 uniqueness

I calculated r(D1,D2)=0.384 and wondered if my grader would consider D2 a proxy for D1. My question: "My correlation matrix has r(D1,D2)=0.384. My grader may think D2 is redundant, as complex tasks take more time. How do I answer this properly?" VIF and partial correlations were new to me. I calculated the cell using numpy.linalg.inv, checked VIF(D2)=1.25 and D2↔D3 partial r=−0.008, then finally reported the 80.3% unique variance after confirming this was correct.

Decision 4 — Fixing the classification_report label swap

I ran the baseline and got 742 as the support under "Silent Fast-Track" but the number of test samples for SFT is 3,955. My prompt: "I have 3,955 test samples for Silent Fast-Track (SFT) but my classification_report has support=742 under 'Silent Fast-Track'. These look like they are swapped. What is going on here?" ChatGPT diagnosed the sklearn alphabetical sorting bug (D→R→S) with target_names but without labels=. I checked the fix with Counter(y_test). Without this bug fix, DM would have been reported with F1=0.889 instead of SFT - a complete misinterpretation.

Decision 5 — Adding SBERT+LR as Model 3 after LightGBM failure

My result RHE recall = 1.9% from SBERT+LightGBM did not make sense. My question: "My complex model has RHE recall=0.019 - almost zero. But my baseline LR gets 0.443. How come LightGBM is_unbalance=True is so bad?" ChatGPT described gradient boosting's tendency to converge to majority in 384-dim space. I implemented and executed Model 3 (SBERT+LR) to check whether the bottleneck was embedding or classifier. SBERT+LR brought the RHE recall to 61.3% - the bottleneck was LightGBM's gradient, not SBERT's embedding.

The Most Meaningful Iterative Refinement

First AI conversation - what ChatGPT said: In my first conversation with the AI about designing dimensions, I explained that I planned to use CDI as D2, adopting the A2 metric. ChatGPT agreed with this plan and helped me translate the aggregation code in pandas to calculate CDI from pr_reviews. This code executed without errors and I saw the D2_cognitive column in my dataframe. Everything seemed to be fine.

What was wrong: When I looked at the summary statistics of D2_cognitive I saw the following: median = 0.000, Q1 = 0.000, Q3 = 0.000. This feature has more than 75% of its values at zero. With the help of ChatGPT I had the correct formula for CDI, assuming that most PRs have reviews. This is not the case in the AIDev-pop subset: most of the agentic PRs (in particular the 18,004 OpenAI Codex PRs) are merged in stealth automation and have no records in pr_reviews, so CDI is not defined and is thus zero. ChatGPT did not tell me about this distribution, because I didn't provide the statistics. I regrettably did not look at the statistics at first.

The revision: I asked ChatGPT to help me with the diagnostic output and the problem. It suggested using body word count. I did not immediately accept this suggestion - I undertook three checks: (1) confirmed that the clusters have a significant spread (SFT=28, DM=110, RHE=302 - a 10× span); (2) calculated VIF=1.25 confirming D2 is the statistically most independent dimension in the feature set; (3) independently thought up the temporal orthogonality argument - that D2 is the only pre-review dimension, while D1, D3 and D4 are realised during or after review. This idea was not given by ChatGPT in its recommendation; I came up with this during the writing of the justification.

Why this refinement is significant: The initial version of this pipeline was a complete - the code was clean, the feature was calculated, and the dataframe was prepared for K-Means. This problem was not visible by only looking at the statistics. To fix it, I had to: (a) independently identify a dataset-specific characteristic unknown to ChatGPT; (b) return and provide a more specific problem statement; (c) check the replacement with three independent tests; (d) independently come up with a new domain argument (temporal orthogonality of pre-review and post-review dimensions) to better motivate it. The D2 is an improvement over the hypothetical CDI, because the same feature (description length) used for clustering in Part 1 is observable by the text classifier in Part 2, thus connecting the two parts of the analysis. Overall: ChatGPT improved the pipeline measurably (VIF proof, k=3 selection, Model 3) but introduced two errors (CDI degeneracy, label swap) requiring independent verification. Core findings hold.